What harmonic phonon calculations capture — and miss — about thermal conductivity

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Analytical and machine-learning models predict lattice thermal conductivity from harmonic phonon properties with useful accuracy, and adding anharmonic descriptors is known to improve them; what the residual of a purely harmonic model contains, and how much of it is label noise, is quantified here on a large set. Harmonic force constants predict the lattice thermal conductivity (PhononDB, phono3py RTA, 300 K) of 749 crystalline solids spanning 4.7 orders of magnitude with cross-validated $R^2 = 0.86$–0.89, using 14 standard phonon descriptors; the plateau persists across linear, ensemble and neural-network models and with 17 additional engineered features (0.89–0.91). On a 229-material subset the residual correlates with a transport-weighted cubic-coupling proxy $|V_3|^2 \equiv \Gamma/W$. Adding anharmonic information raises R^2 on the subset from 0.80 by +0.05 (the scalar proxy) and +0.11 (a coarse RTA κ); both are computed on the label's own force-constant route, so the gain is not purely anharmonic. Label noise, estimated against an independent database, caps R^2 at ≈ 0.96 (95% bootstrap interval 0.93–0.98) if both databases are equally noisy, above the plateau: label noise alone does not explain it. Four rigid oxides and eight layered compounds deviate in $|V_3|^2$.

I. INTRODUCTION

Lattice thermal conductivity κ is a central quantity in thermal management of electronics [1], thermoelectric energy conversion [2], and thermal barrier coating design [3]. First-principles prediction of κ through the phonon Boltzmann transport equation (BTE) with anharmonic (third-order) force constants is now well established [4, 5] and agrees well with experiment for many crystalline solids [6]. However, the computational cost of obtaining third-order force constants remains substantial, since the number of displaced-supercell calculations grows quickly with the supercell size and the interaction range, which limits the throughput of BTE-based screening [7].

Harmonic (second-order) force constants are considerably cheaper to obtain, requiring only a single set of displaced-atom calculations. From the harmonic dynamical matrix one can extract phonon dispersion relations, sound velocities, Debye temperatures, phonon density of states, and derived quantities such as the three-phonon phase space [8, 9]. Analytical models built on such quantities [10, 11] and machine-learning models [12–14] estimate κ with an accuracy that depends on the feature set, model complexity and dataset; the most accurate of them add an anharmonic ingredient beyond harmonic information — a reduced third-order descriptor [12], a Grüneisen parameter [13], or an anharmonicity measure [14, 15]. What has received less attention is the characterization of the residual of a purely harmonic model: what physical information it lacks, how much of it is label noise, and whether any inexpensive descriptor can recover it. If the variance left below the plateau reflects missing physics — specifically information contained in higher-order force constants — then improving harmonic models is fundamentally limited.

We address this question on 749 materials from PhononDB [16] for which both harmonic force constants and phono3py κ at 300 K (relaxation-time approximation, RTA) are available. We train models of increasing complexity, characterize the plateau, and correlate the harmonic residual with anharmonic quantities available for a 229-material subset. Consistent with that literature, the study quantifies two things: (i) The harmonic plateau is placed against a label-noise bound measured between two independent first-principles databases, and lies below its central estimate. (ii) The residual of the harmonic model is correlated with a cubic-coupling proxy, and the part of that proxy which harmonic descriptors do not predict raises R^2 by 0.05 on the 229-material subset.

II. DATA AND METHODS

A. Dataset

We use PhononDB [16] materials that have both harmonic force constants (FORCE_SETS from phonopy [17]) and lattice thermal conductivity at 300 K computed by phono3py [5] in the relaxation-time approximation (RTA; every PhononDB calculation log we inspected runs conductivity-RTA, with isotope scattering and the tetrahedron method), using the trace average $(\kappa_{xx} + \kappa_{yy} + \kappa_{zz})/3$. PhononDB computes these κ from second- and third-order force constants evaluated with a polynomial machine-learning potential (polymlp) trained on DFT (phonopy 2.47.1,

phono3py 3.23.0), while our harmonic descriptors come from PhononDB's separate DFT FORCE_SETS: the label and the descriptors rest on different second-order force constants. Of 766 κ records, two have no harmonic record ($NaLa_2RuO_6$, Rb_2H_6Pt), three are excluded as κ convergence artifacts (BaO_2, $Pr_2C(NO)_2$, Na_2PdC_2): the same RTA calculation on the $6 \times 6 \times 6$ mesh of the anharmonic subset diverges for each (κ not finite), BaO_2's κ_{zz} is 1.3×10^4 W/mK (Phonix's value diverges too), and Phonix's independent values for the other two are 19× and 46× lower. They are also the three largest residuals of the model, so their exclusion is reported as a sensitivity: with them, Ridge R^2 is 0.82. Six other materials whose $6 \times 6 \times 6$ run diverges keep their labels (2–4 W/mK for four of them, 13 W/mK for LiC, 102 W/mK for BeSe), within a factor of two of Phonix where it has them, and stay. Two are excluded because their κ record and their harmonic record belong to different structures (K_2O, $Al(OH)_3$). Ten more are excluded because the electronegativity-contrast feature needs at least two distinct elements with a Pauling electronegativity in the table we use, which omits Hg, Yb and Pa: this excludes the elemental solids B and Si (two polymorphs), three Hg compounds and four Yb compounds, leaving 749; the exclusion of materials with unresolved sound velocities removes none. In the 20 retained materials that contain Hg, Yb or Pa, those elements are left out of the contrast. The dataset spans six crystal systems: orthorhombic ($N = 208$), monoclinic (174), cubic (136), trigonal (104), tetragonal (85), and hexagonal (42). The DFT harmonic records of 78 of the 749 have imaginary modes on the $12 \times 12 \times 12$ mesh (46 of them at 100 or more mode points); without these 78, Ridge R^2 is 0.87.

B. Harmonic features

From the harmonic force constants of each material we extract 14 features with phonopy 3.4.0:

- **Sound velocities** (3): longitudinal v_{LA}, transverse v_{TA}, and sv2, the Brillouin-zone-weighted mean of the squared group velocity per mode ($12 \times 12 \times 12$ mesh; modes below 0.1 THz set to zero but counted among the $3N$ modes), which captures the full dispersion rather than only the long-wavelength limit.
- **Debye temperature** (1): θ_D computed from the second moment of the phonon density of states.
- **Phase space** (1): W, the Brillouin-zone mean of the two-phonon density of states $D_2(\omega) + D_2^-(\omega)$ evaluated at each mode's frequency, built from the one-phonon density of states with energy conservation only (no momentum conservation, no occupation or $1/(\omega\omega'\omega'')$ factors; cf. the three-phonon phase space of [8] and its weighted form [9]), computable from the harmonic dispersion alone. The one-phonon density of states is a 200-bin histogram of the irreducible-wedge frequencies without their multiplicity weights.
- **Spectral shape** (5): spectral spread (the standard deviation over the mean of the irreducible-wedge frequencies, unweighted), the mean largest adjacent-branch gap (at each $\mathbf{q}$, the largest spacing between adjacent branches, averaged over the zone; positive also for gapless spectra), the fraction of sv2 below the median frequency, the spectral entropy of the sv2-weighted frequency distribution ($\min(20, 3N)$ bins, normalized by the logarithm of the bin count), and the DOS width ratio (the weighted standard deviation of the frequencies over the maximum frequency).
- **Structural** (4): average atomic mass, electronegativity contrast (maximum minus minimum Pauling electronegativity over the atoms with a value in the table), $f_{\mathrm{c,mean}}$ (the mean Frobenius norm of the 3×3 force-constant blocks between each primitive-cell atom and every supercell atom, the on-site block included, in eV/Å^2), and interaction range (the fraction of off-site blocks with a norm above 0.01 eV/Å^2; 1.0 for 77% of the materials). Both force-constant features depend on the supercell PhononDB used.

All 14 features are computable from FC2 and the composition in seconds per material. They enter the models as $\log_{10}$ of v_{LA}, v_{TA}, sv2, the mass and W, and linearly otherwise (θ_D, the five spectral-shape features, the electronegativity contrast, $f_{\mathrm{c,mean}}$ and the interaction range), standardized inside the cross-validation folds for Ridge and the network. The target variable is $\log_{10}\kappa$, with κ the PhononDB RTA trace average as shipped (rounded to 0.01 or 0.001 W/mK). As a robustness check, we verify that adding 17 engineered features (five Morse-potential estimates, six harmonic proxies on an $8 \times 8 \times 8$ mesh — inverse group velocities, frequency skewness and the mode count — and six crystal-system indicators) yields minimal improvement in nonlinear models (Sec. III A).

C. Models

We train three model families with scikit-learn 1.9.0: (1) Ridge regression ($\alpha = 1$, linear baseline), (2) gradient boosting (500 trees, max depth 4, other settings default), and (3) a neural network (MLPRegressor, layers 256-128-64,

up to 1000 iterations), each with random state 42. All are evaluated by 10-fold cross-validation repeated over ten random fold assignments, and every cross-validated number is the mean over the ten assignments, except the grouped cross-validation, which uses one assignment per grouping. R^2 is the mean over the folds for the model comparison, the ablation, the learning curves and the oracle tests, and the pooled out-of-fold R^2 for the grouped cross-validation, the error by κ range and the Slack baseline; the $\pm$ is the standard deviation over all 100 folds, a spread between folds rather than an uncertainty of the mean (the ten assignments' means of the Ridge R^2 lie within 0.01 of each other). The nearest-neighbour estimate uses no cross-validation. Feature ablation adds feature groups incrementally: sound velocities → Debye temperature → spectral shape → structural descriptors → phase space W.

D. Anharmonic data

Third-order force constants were available to us for 247 PhononDB materials; on these we ran our own phono3py RTA calculations, which stay below 10^3 W/mK for 238 of them (a threshold that removes the diverged runs), and 229 of these overlap with the main dataset after exclusion criteria. From these runs (phono3py 4.8.2 [5] with its default options: second- and third-order force constants computed by finite differences from PhononDB's displacement–force datasets, a fixed $6 \times 6 \times 6$ mesh, 300 K, no isotope scattering) we extract the BZ-averaged phonon scattering rate Γ, weighted by each mode's squared group velocity ($v_{\mathbf{q}}^2 / \sum v_{\mathbf{q}}^2$). These runs differ from PhononDB's κ by 0.19 dex (standard deviation of $\log_{10}$), which attenuates correlations with Γ. Two settings cause it, with opposite signs: the coarse mesh lowers κ in most materials, and the absence of isotope scattering raises it strongly in a few, where some modes barely scatter at $6 \times 6 \times 6$ (SrS_2: 477 W/mK against a label of 3.35 W/mK; without SrS_2 the difference is 0.13 dex). We define the effective cubic coupling strength as $|V_3|^2 \equiv \Gamma / W$, which divides out the energy-conserving phase space W. It is a proxy, not a direct FC3 norm: Γ is transport-weighted while W is a zone mean, and W omits the momentum-conservation, occupation and $1/(\omega\omega'\omega'')$ factors that Γ contains, so part of what harmonic descriptors predict of $|V_3|^2$ is harmonic by construction. Γ and our RTA κ also rest on the force constants of the label's own route (PhononDB's displacement–force datasets), while the descriptors come from the separate DFT FORCE_SETS (Sec. II A), so what they add to the harmonic models (Sec. III E) includes harmonic content of the label's route as well as anharmonic content. The v^2 weighting of Γ captures the component of scattering relevant for thermal transport. These quantities are used for the residual analysis and the oracle tests (Sec. III E), never as features of the harmonic models. Γ is in THz; $|V_3|^2$ is read on a logarithmic scale, in arbitrary units.

III. RESULTS

A. Harmonic prediction accuracy

With 14 features, Ridge regression achieves $R^2 = 0.86 \pm 0.03$, gradient boosting $R^2 = 0.88 \pm 0.03$, and the neural network $R^2 = 0.89 \pm 0.03$ on 749 materials spanning κ from 0.01 to 449 W/mK (Fig. 1). Feature ablation on Ridge reveals a clear hierarchy: sound velocities alone yield $R^2 = 0.61$; adding Debye temperature gives 0.64; spectral shape brings 0.79; structural descriptors reach 0.84; and including the phase space W completes the set at 0.86. The marginal contribution of W (+0.02) is context-dependent, consistent with its role as a kinematic complement to sound velocity.

Extending to 31 descriptors (+17 engineered features), Ridge improves to $R^2 = 0.89$ and the neural network to $R^2 = 0.91$, the network gaining ≈ 0.02 over the 14-feature baseline. Ridge is saturated by ≈ 400 training materials (gain < 0.01), while gradient boosting still gains 0.03 between ≈ 400 and ≈ 670; these gains compare training on a random 60% of each 10-fold training fold (≈ 400 materials) with training on all of it, tested on the same held-out fold (Fig. 5 shows the curve against the size N of random subsets, with five-fold cross-validation inside each): the linear plateau is not data-limited, the nonlinear one may not be fully saturated.

Three robustness checks bound the plateau. (i) Grouping the cross-validation folds by composition, chemical system or structural prototype (space group and stoichiometry) changes R^2 by at most 0.02 (Ridge 0.86, gradient boosting 0.87–0.88), so polymorph leakage does not inflate it. (ii) Label noise. On the 571 materials shared with Phonix [18] (its entries averaged per material in $\log_{10}\kappa$), an independent first-principles database (VASP and ALAMODE force constants, Peierls–Boltzmann RTA), $\log_{10}\kappa$ differs by 0.208 dex (standard deviation; 0.078 dex robust, from the median absolute deviation, so the spread is dominated by a tail of large disagreements). With sd $\log_{10}\kappa = 0.706$ on the 749 materials, label noise σ bounds R^2 at $1 - (\sigma/0.706)^2$: 0.99 for the robust 0.078 dex, 0.96 if the 0.208 dex is split equally between the two databases (0.147 dex each), and 0.91 if all of it is PhononDB's. The plateau (0.86–0.89) lies 0.07–0.10 below the central bound, whose 95% bootstrap interval over the 571 materials is 0.93–0.98: label noise alone does not explain it unless nearly all of the disagreement is PhononDB's, in which case the cap (0.91) lies close

above the plateau. A second route estimates PhononDB's own share: on the same 571 materials, the covariance of the cross-validated residual with the PhononDB − Phonix difference equals the variance of PhononDB's label error if the model's error is independent of both databases' errors and the two databases' errors are uncorrelated. It gives ≈ 0.093 dex (Ridge; 0.050 dex with gradient boosting), a cap of $R^2 \approx 0.98$: on those assumptions, label noise explains little of the plateau. Separately, among the 592 unique nearest-neighbour pairs from different structural prototypes, the closest 5–20% differ by 0.17–0.20 dex (rms/$\sqrt{2}$): a local estimate, in the densest part of descriptor space, of the error that label noise and missing physics leave (plus the residual distance between the pair). It is not converted to R^2, because the close pairs sample only that part of the space. (iii) R^2 is a range statistic: the root-mean-square error is 0.24 dex (gradient boosting) to 0.26 dex (Ridge) overall; for gradient boosting R^2 is 0.77 on linear rather than logarithmic κ, and within 1–10 W/mK (56% of the materials) R^2 is 0.35 at an error of 0.21 dex. A Slack-type baseline, $\kappa \propto \bar{M}\theta_a^3 \delta\, n^{-2/3}$ with the acoustic Debye temperature $\theta_a = \theta_D n^{-1/3}$ (so $\log \bar{M} + 3\log\theta_D + \log\delta - \frac{5}{3}\log n$, γ held constant), reaches only $R^2 = 0.40$ with Slack's exponents, 0.48 when one overall scale is fitted to them, and 0.59 when every exponent is fitted.

B. Characterizing the residual

To determine whether the harmonic residual contains anharmonic information, we correlate it with the anharmonic scattering rate on the 229-material RTA overlap; the residual is that of a model cross-validated on these 229 materials.

Using a 5-feature baseline (sv2, mass, $f_{\mathrm{c,mean}}$, θ_D, W), the residual correlates with $\log(\Gamma)$ at $r = -0.54$. The full 14-feature model yields $r = -0.28$ ($p < 10^{-4}$): the reduction is the part of Γ that the harmonic descriptors already predict. The 14-feature residual correlates with $\log|V_3|^2$ at $r = -0.27$ ($p < 10^{-4}$) (Fig. 2).

On all 749 materials, the residual of the 14-feature model shows at most a weak dependence on material class (ANOVA $p = 0.04$–0.07 over the fold assignments, across the nine dominant-element groups with at least 15 materials, the dominant element being the one with the largest stoichiometric count, ties going to the element listed first in the formula) and no serial structure when the residuals are ordered by predicted κ or longitudinal sound velocity (Durbin–Watson 1.97–2.04, a check for unmodelled trends along those variables). Ordered by mass, Durbin–Watson is 1.77 (permutation $p \leq 0.002$ on every fold assignment), from polymorphs of one composition, which share a mass and sit side by side: with one residual per composition it is 1.91 ($p = 0.11$).

C. Bond stiffness and the coupling proxy

The correlation between $\log(f_{\mathrm{c,mean}})$ and $\log(|V_3|^2)$ is $r = +0.10$ ($p = 0.15$, $n = 229$). Controlling for atomic mass gives partial $r = -0.05$ ($p = 0.48$). A Spearman rank correlation gives $\rho = +0.17$ ($p = 0.01$) before mass control and $\rho = +0.04$ ($p = 0.54$) after. At this sample size, $|r| > 0.13$ is the 95% significance threshold (≈ 0.18 for 80% power); the 95% confidence interval of the partial r is -0.18 to $+0.08$ (Fig. 3). Because $|V_3|^2 \equiv \Gamma/W$ contains the harmonic stiffness and the mass by construction (Sec. II D), this near-zero partial correlation does not show that the cubic and harmonic force constants are independent.

Controlling for sv2 and θ_D in addition to mass, the partial correlation of $f_{\mathrm{c,mean}}$ with $|V_3|^2$ is $r = -0.28$ ($p < 10^{-4}$). This reflects redundancy within the harmonic descriptor space rather than access to anharmonic information: removing $f_{\mathrm{c,mean}}$ from the 14-feature model changes the residual's correlation with $|V_3|^2$ by 0.007.

D. Material classes at the boundary

Two material classes deviate in $|V_3|^2$ (Fig. 4). These examples are illustrative rather than statistically representative, and their p-values are exploratory (not corrected for the multiple comparisons in this section).

Rigid oxides (TiO_2 and SnO_2 in the rutile structure, cubic ReO_3, tetragonal ZrO_2 ($P4_2/nmc$); $n = 4$): high bond stiffness and high $|V_3|^2$ (1.36 standard deviations above mean, one-sided Mann–Whitney $p = 0.006$ against the 217 subset materials outside both classes). The small sample size limits statistical power. One hypothesis is soft tilting modes of the metal–oxygen framework (ReO_3's harmonic record has imaginary modes), which would raise Γ/W through its $1/\omega$ and occupation factors as well as through the cubic coupling; it is not tested here. The harmonic model (Ridge, cross-validated on the 229) overpredicts κ for three of the four, by 0.7–1.3 dex (SnO_2's residual is ≈ 0).

Layered compounds (GaSe ($P\bar{6}m2$ and $P6_3/mmc$), GaS, GaTe, WS_2, MoS_2, WSe_2, BiI; $n = 8$): low $|V_3|^2$ (1.60 standard deviations below mean, $p < 0.001$). One hypothesis is covalent, symmetric in-plane bonding; another is a restricted three-phonon phase space, which W does not see (it omits momentum conservation) and which lowers Γ/W

with no change in the cubic coupling. Neither is tested here. The harmonic model underpredicts κ for six of the eight, by 0.13–0.46 dex (WS_2's residual is ≈ 0, BiI is overpredicted by 0.19 dex).

Both deviations survive mass control (rigid oxide $p = 0.005$, layered $p < 0.001$). The named lists were matched to the materials present in the RTA subset; classes defined by structural rule instead (stiff cubic, tetragonal or orthorhombic cells of 3–12 atoms with $f_{c,\mathrm{mean}}$ above the 75th percentile of the 749, $n = 8$; hexagonal or trigonal cells with short interaction range, $n = 18$) give $p = 0.009$ and $p = 0.0001$, but they share members with the named lists (all four oxides, seven of the eight layered materials); without those members, $p = 0.19$ ($n = 4$) and $p = 0.08$ ($n = 11$). Bootstrap 95% intervals on the z-scores (2000 resamples of the subset, those with fewer than two class members discarded) are -1.84 to -1.41 (layered, $n = 8$) and $+0.60$ to $+2.22$ (rigid oxides, $n = 4$). Averaging the tetrahedron integration weights over degenerate bands changes the eight layered materials' transport-weighted Γ by at most 0.002 dex, so the layered offset does not come from the degenerate-band treatment.

E. Oracle tests on the harmonic plateau

Adding the $|V_3|^2 \equiv \Gamma/W$ scalar as an oracle (its part the 14 features do not predict: the out-of-fold residual of a Ridge model, fitted inside each training fold) improves R^2 by $+0.05$ (from 0.80 to 0.85 on the 229-material subset). This scalar does not exhaust what the label's route adds: our coarse-mesh RTA κ itself, added to the 14 features, raises R^2 on the same subset from 0.80 to 0.91 ($+0.11$, same protocol), and alone reaches 0.90. Part of the gap above the harmonic plateau (0.80 to 0.91 on the subset) is closed by third-order information computed explicitly, together with the label's own harmonic route that Γ and the RTA κ share (Sec. II D).

IV. DISCUSSION

A. Interpretation

The kinetic expression $\kappa = \frac{1}{3}\sum_{\mathbf{q}} C(\mathbf{q})\, v^2(\mathbf{q})\, \tau(\mathbf{q})$ separates into the harmonic quantities C and v and the scattering time τ, which carries the anharmonic coupling. In the BZ-averaged form, this separation can be made explicit in a minimal form:

$$\kappa \approx A \cdot \mathrm{sv2}^a \cdot \Gamma^b, \qquad a = +1,\ b = -1 \text{ in the kinetic expression.} \tag{1}$$

A two-variable regression on the 229-material subset gives in-sample $R^2 = 0.78$ (cross-validated $R^2 = 0.74 \pm 0.14$) with coefficients $a = +1.11$, $b = -0.80$, consistent with the physical expectation ($a \approx +1$, $b \approx -1$). Since Γ carries harmonic factors of its own (Sec. II D), the fit restates the kinetic expression rather than testing a separation of the two orders.

B. Tested descriptors that do not bridge the gap

We evaluated several candidate descriptors: Morse-potential Grüneisen parameter ($\rho = +0.23$ with the 14-feature residual, the sign opposite to a Grüneisen effect, R^2 lift $= +0.008$), inverse group velocity and spectral skewness proxies (not significant), and an element-composition proxy for $|V_3|^2$ (stoichiometric element fractions; cross-validated $R^2 = 0.51$ for $\log|V_3|^2$, κ lift $= +0.001$). None meaningfully reduced the harmonic residual.

Prior work improved κ prediction by adding anharmonic descriptors to harmonic ones [12–14]; the present results quantify the room for such gains: the headroom above the harmonic plateau before the label-noise limit is 0.07–0.10 at the central estimate.

C. Practical implications

For high-throughput screening where ranking matters more than absolute accuracy, harmonic models provide useful κ predictions with median relative error 30–34% (Spearman $\rho = 0.92$–0.93 over all materials, 0.72–0.74 within 1–10 W/mK). For higher precision, anharmonic calculations remain necessary. The $R^2 \approx 0.89$ level offers a practical benchmark: in our tests, richer harmonic descriptors added little (Sec. III A), while third-order information added more (Sec. III E).

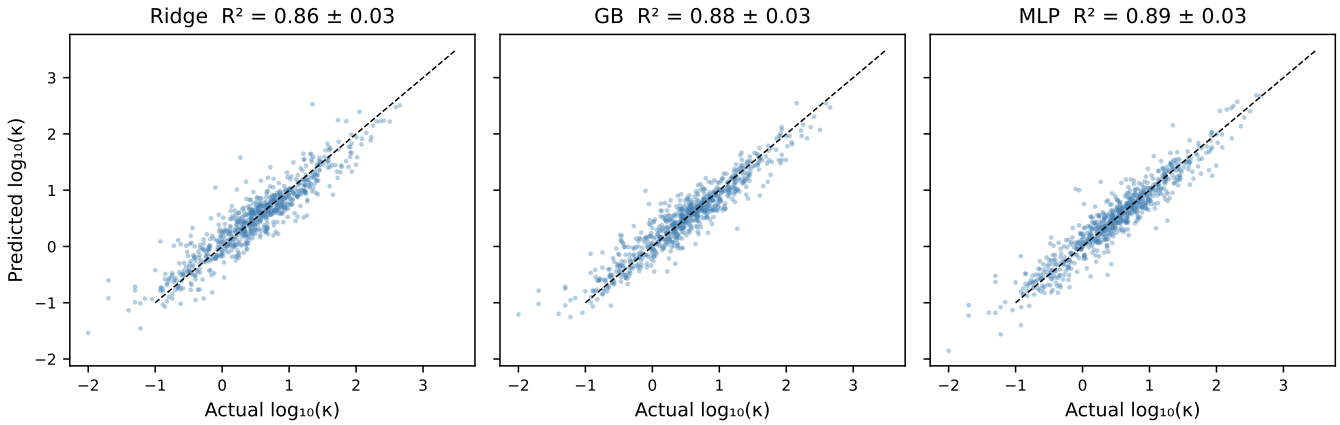


FIG. 1. Predicted vs. actual $\log_{10}(\kappa)$ for Ridge, gradient boosting, and neural network models with 14 harmonic features on 749 materials (out-of-fold predictions averaged over the ten fold assignments).

V. CONCLUSION

Fourteen standard harmonic descriptors predict lattice thermal conductivity with $R^2 = 0.86$–0.89 on 749 crystalline solids spanning 4.7 orders of magnitude in κ. On a 229-material subset, the remaining variance correlates with a cubic-coupling proxy, and oracle tests show that information from the label's own force-constant route, anharmonic and harmonic, closes part of the remaining gap (+0.05 from the scalar Γ/W, +0.11 from a coarse RTA κ); no tested inexpensive descriptor accesses it. These findings help delineate when anharmonic calculations become necessary.

DATA AVAILABILITY

All harmonic features, RTA thermal conductivities, analysis scripts, and 55 reproducibility checks across five test files (ceiling, residual, independence, outliers, robustness) are publicly available at `https://github.com/alex-toader/st_kappa_ceiling` (DOI: 10.5281/zenodo.19570128). The underlying data are from PhononDB [16] and, for the label-noise estimate, Phonix [18].

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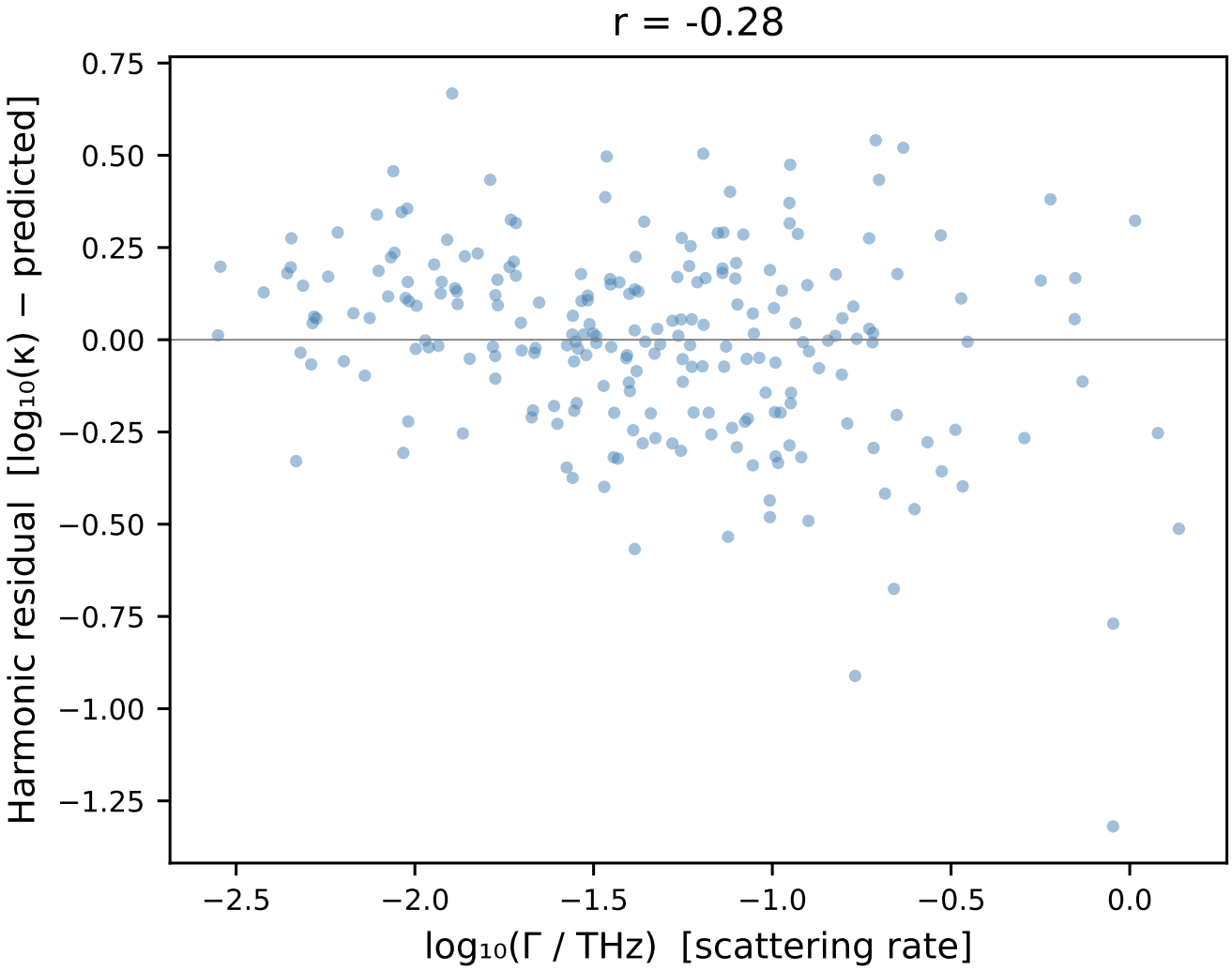


FIG. 2. Cross-validated (out-of-fold) harmonic residual, averaged over the ten fold assignments, vs. anharmonic scattering rate Γ on the 229-material RTA subset.

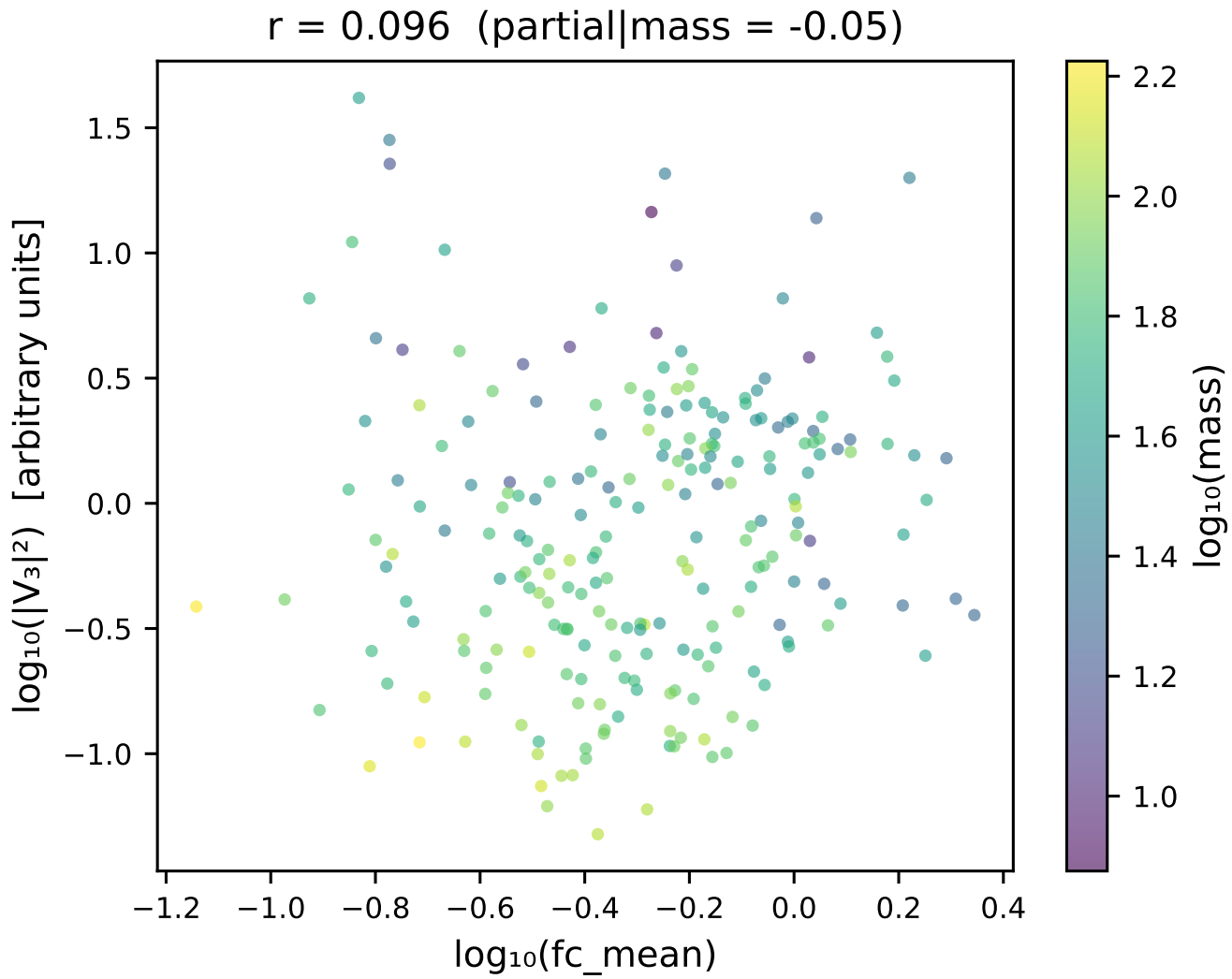


FIG. 3. Bond stiffness ($f_{c,\mathrm{mean}}$) vs. cubic coupling strength ($|V_3|^2$), colored by atomic mass. Neither the raw Pearson correlation ($r = +0.10$, $p = 0.15$) nor the mass-controlled partial correlation ($r = -0.05$) is significant; since $|V_3|^2$ contains stiffness and mass by construction, this does not show that cubic and harmonic force constants are independent (Sec. III C).

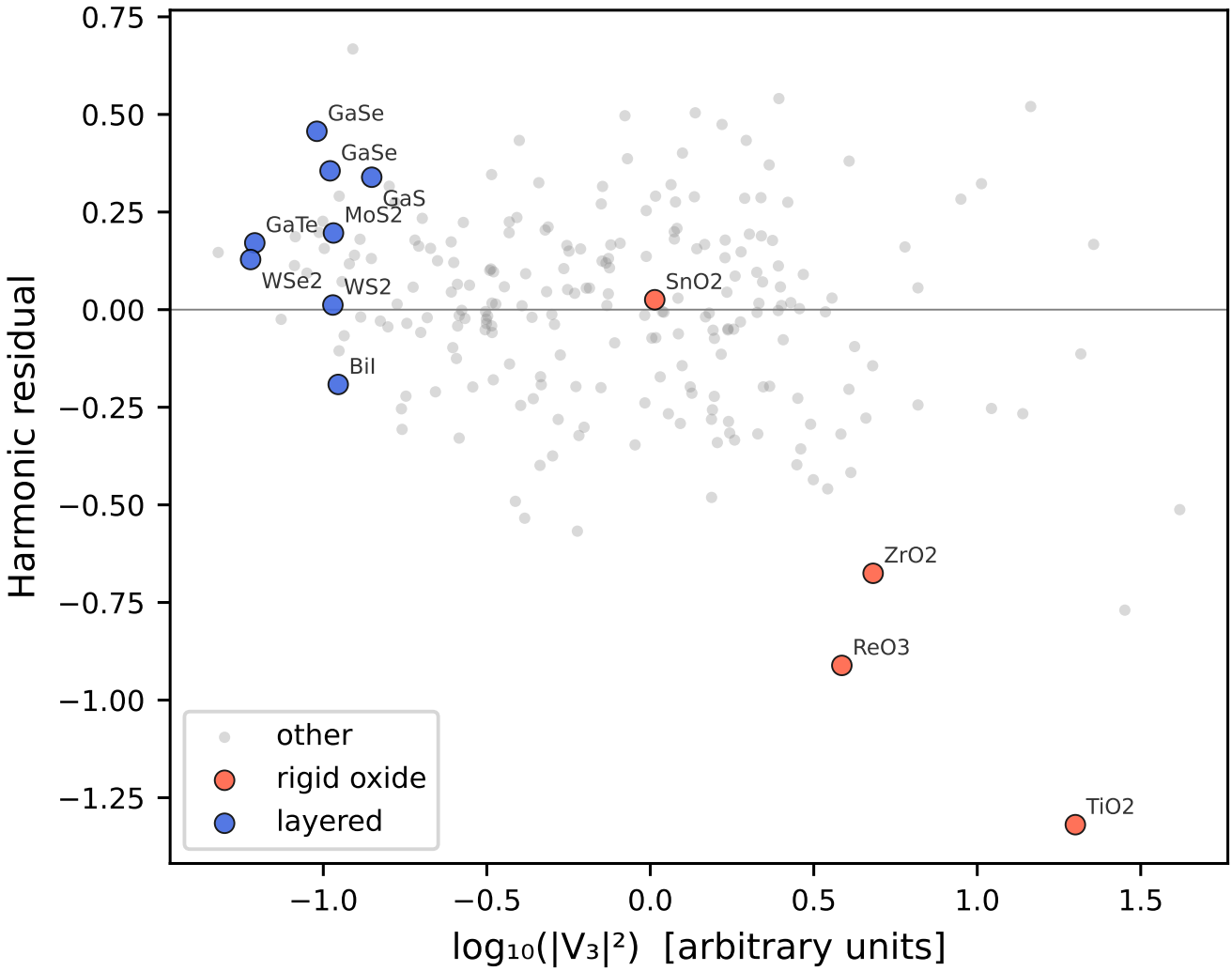


FIG. 4. Harmonic residual (out-of-fold, averaged over the ten fold assignments) vs. $|V_3|^2$ with rigid oxides (red) and layered compounds (blue) highlighted.

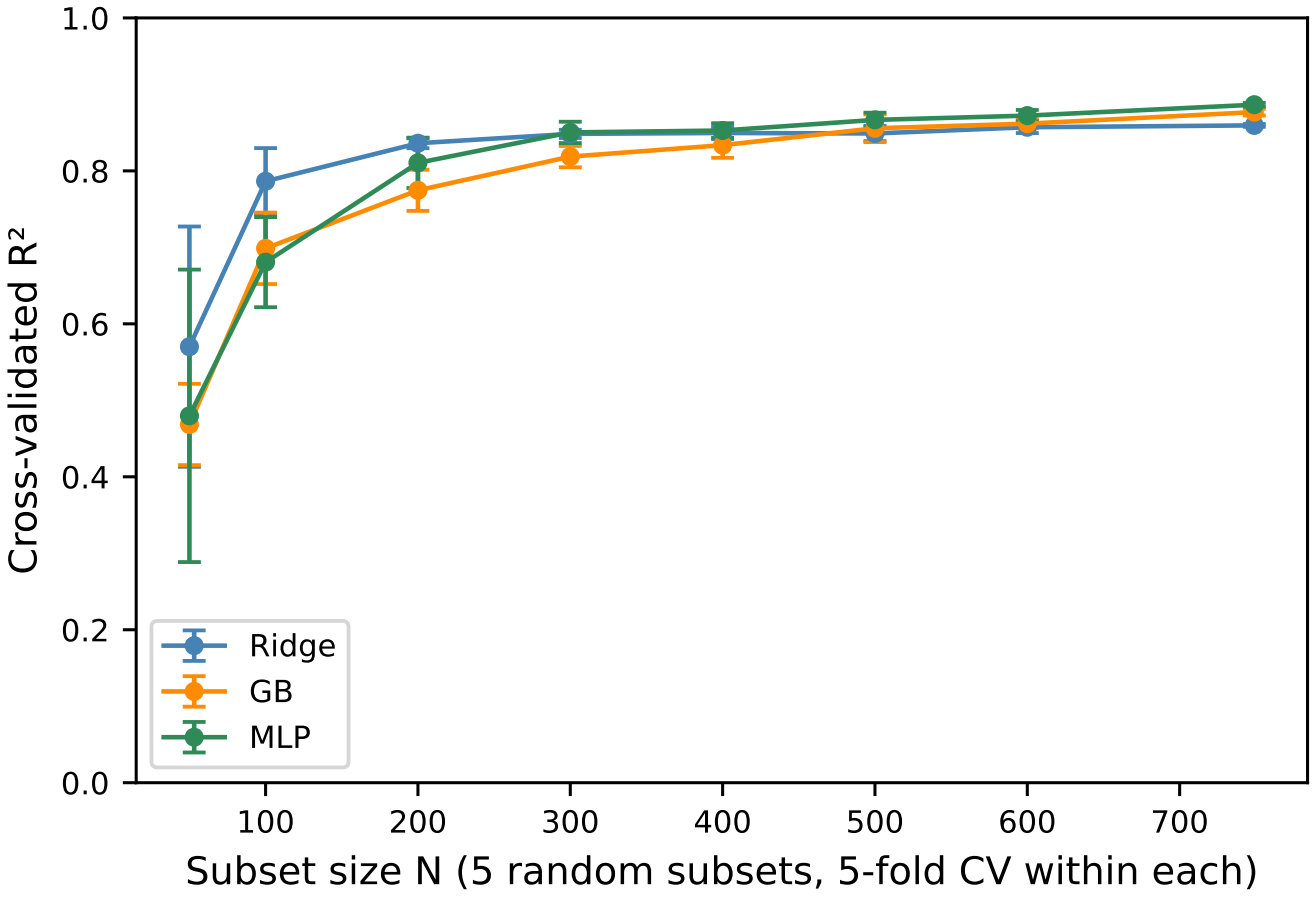


FIG. 5. Cross-validated R^2 against the size N of random subsets (five subsets per size, five-fold cross-validation within each) for three model families. Ridge saturates by $N \approx 400$; the nonlinear models still rise towards the full set (the 10-fold numbers are in §3.1).